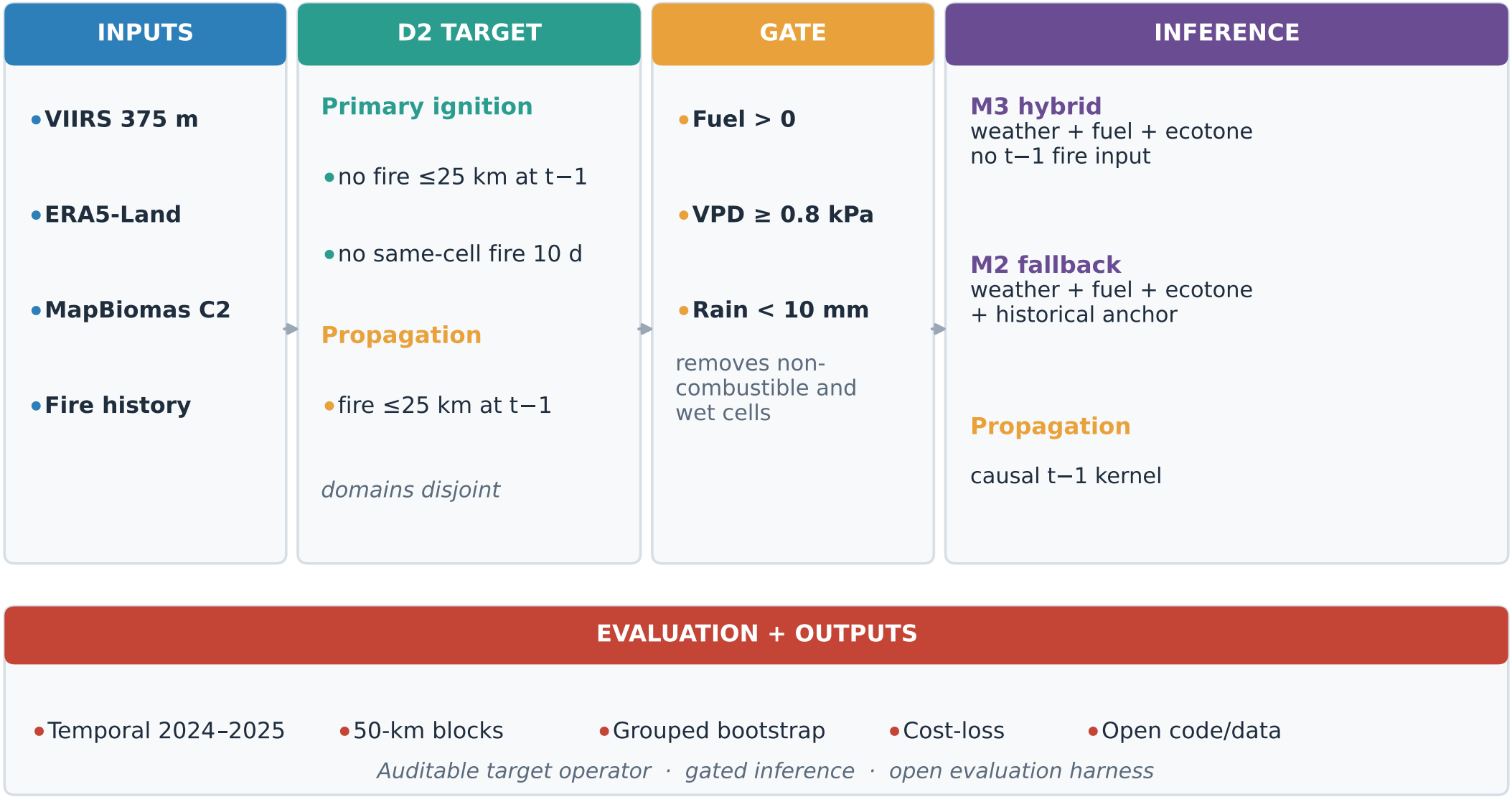


Figura 1. Computational framework: multi-source inputs, D2 target operator, biophysical gate, decoupled M3/M2 ignition and t-1 propagation, reproducible evaluation and open outputs.

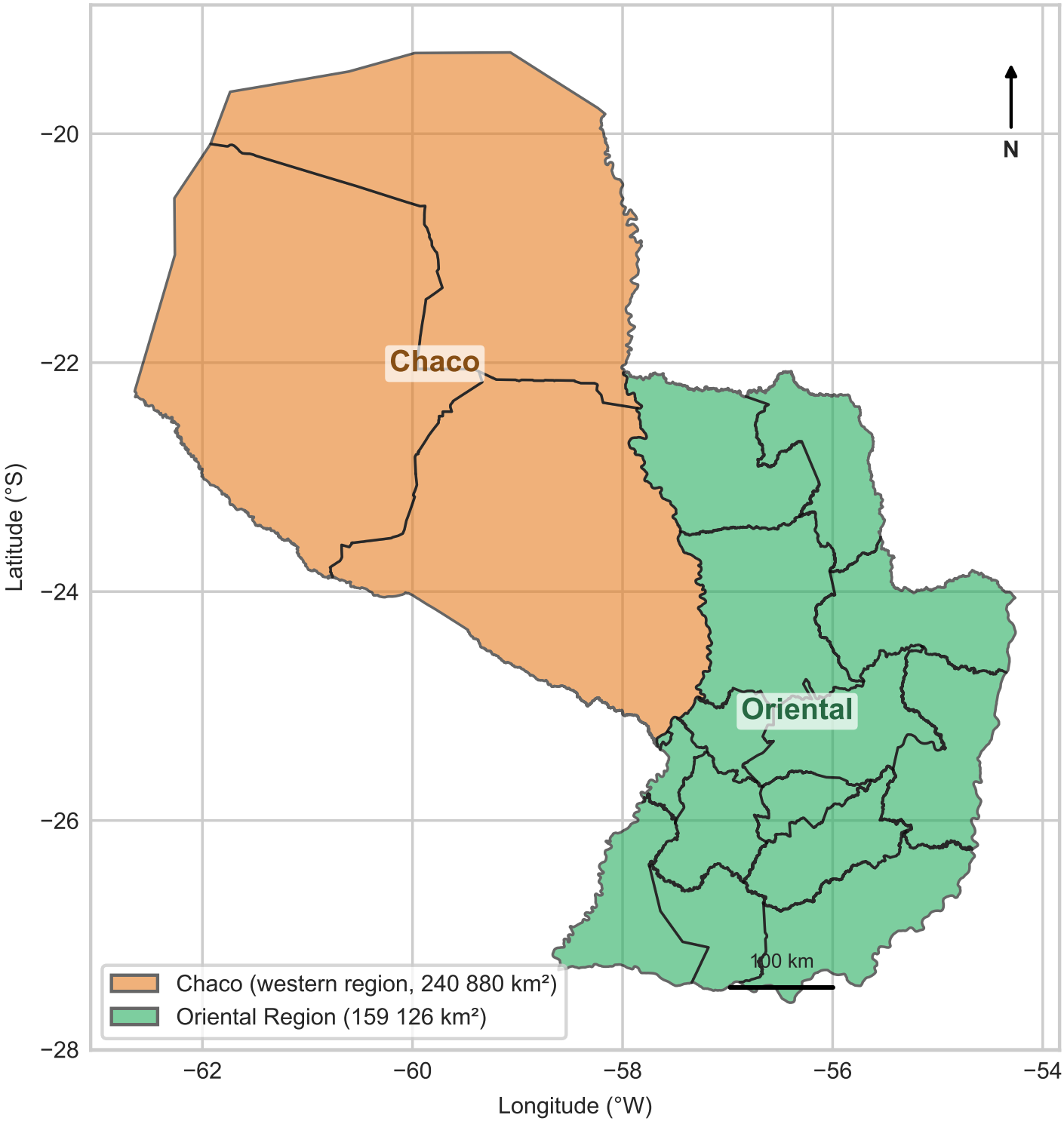


Figura 2. Study domain and regional split: Dry Chaco (west) and Oriental Region (east) over the national 1-km² grid (EPSG:32721).

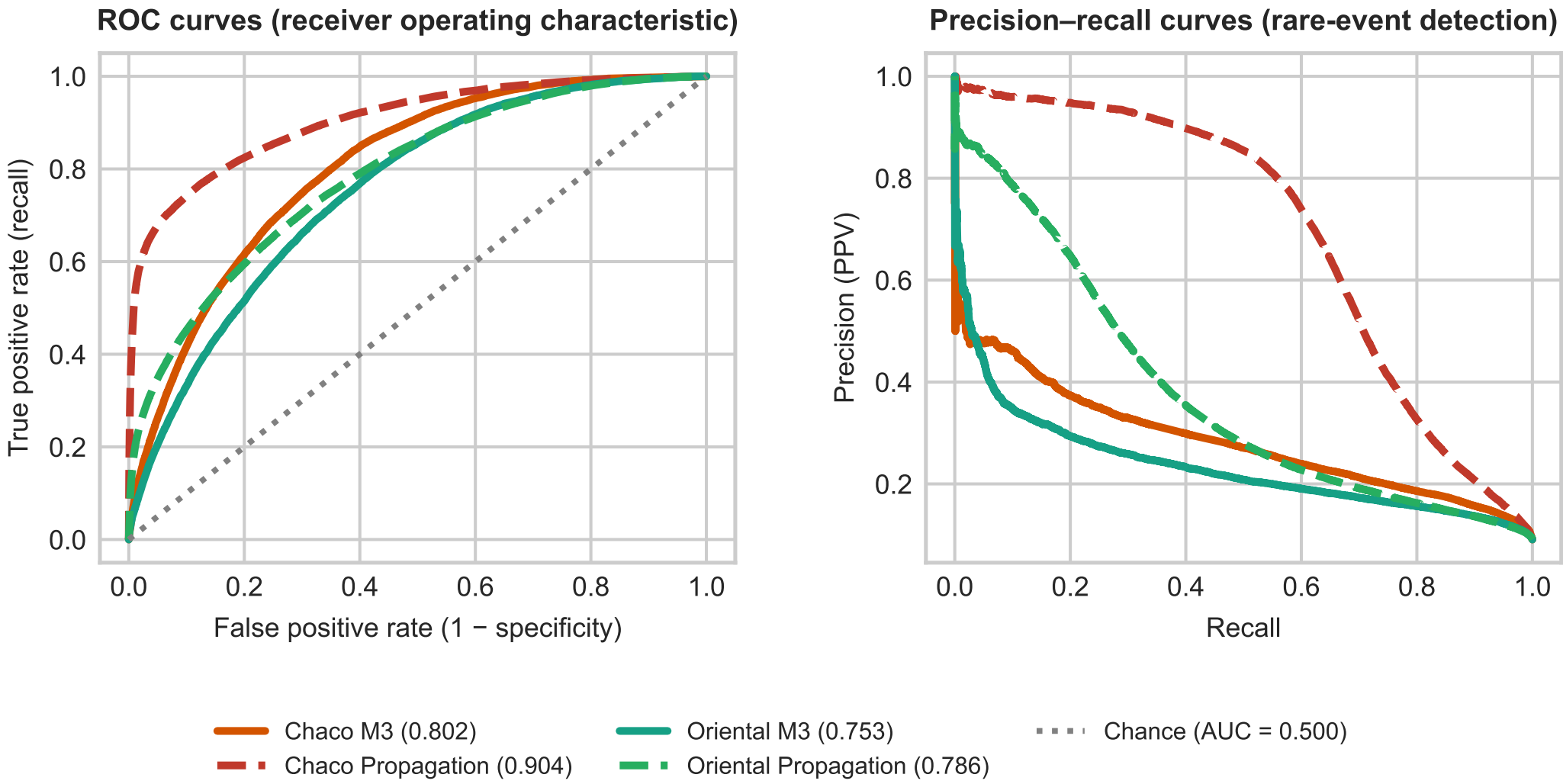


Figura 3. ROC and precision?recall curves for the retrospective D2 test (2024?2025), by region and task. The ignition curves correspond to M3; propagation uses the causal t-1 layer.

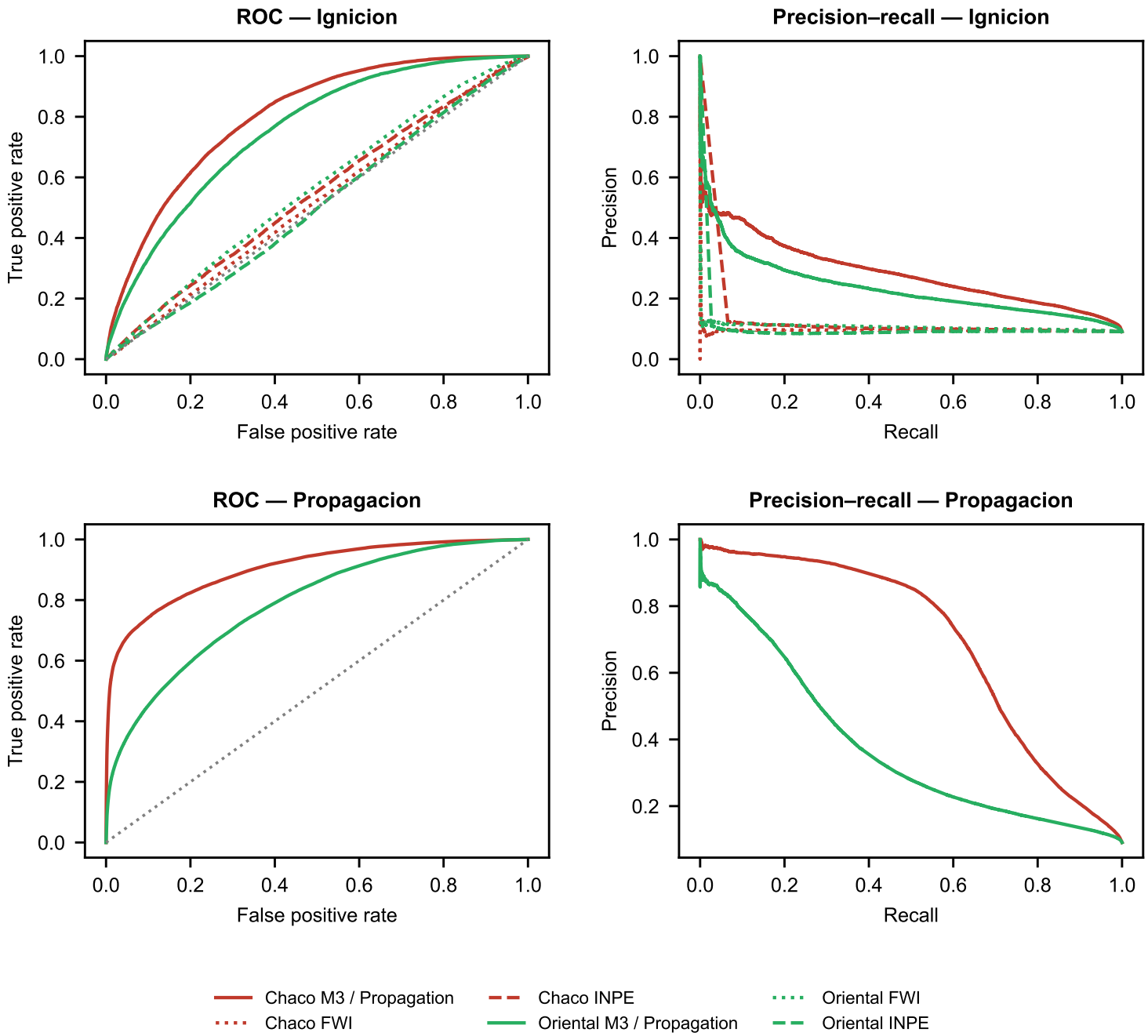


Figura 4. ROC and precision?recall curves for M3 against full Canadian FWI and INPE under D2.

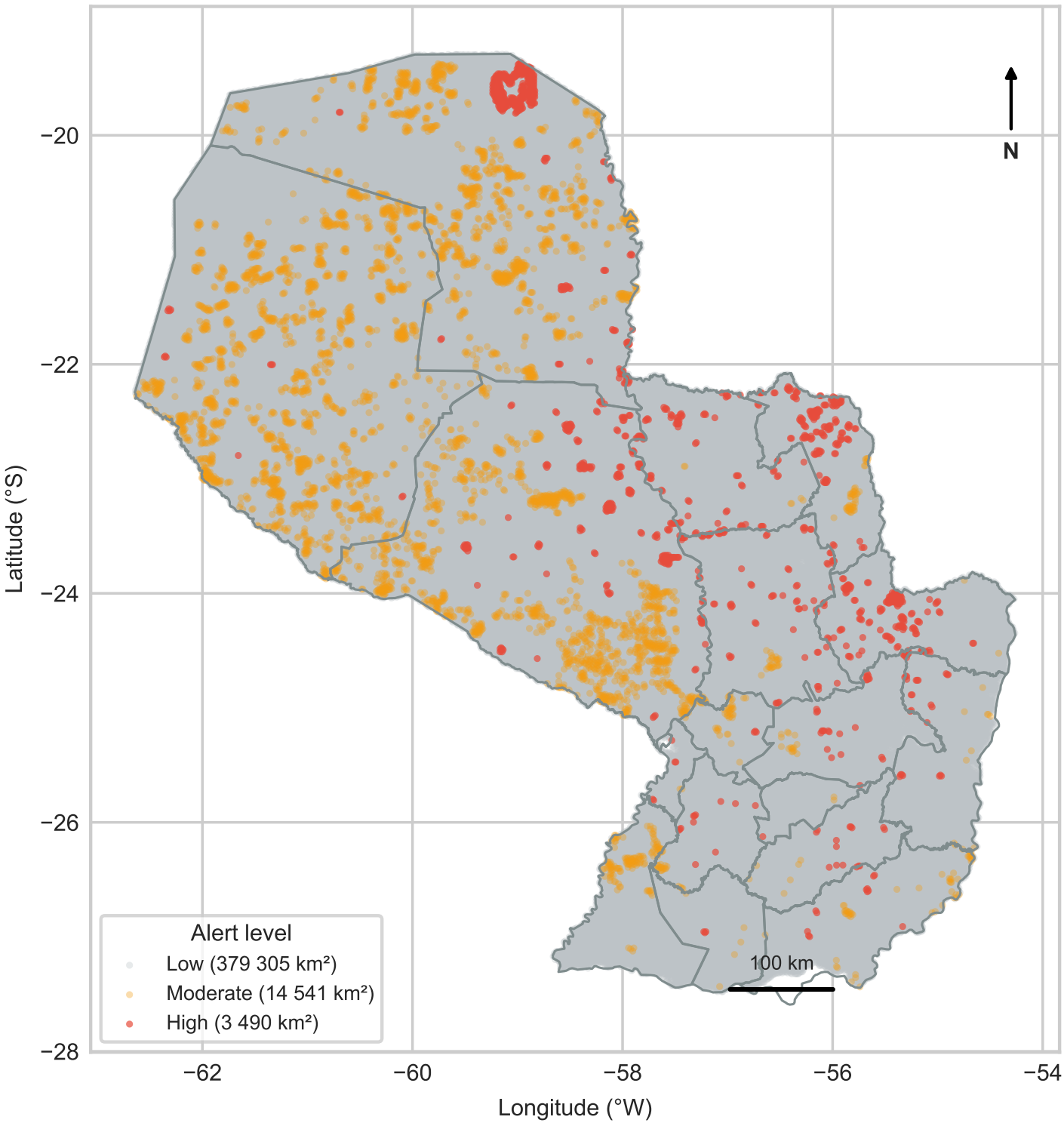


Figura 5. Illustrative national 1-km² alert map for 2024-09-09, showing the spatial distribution of low, moderate and high alert cells.

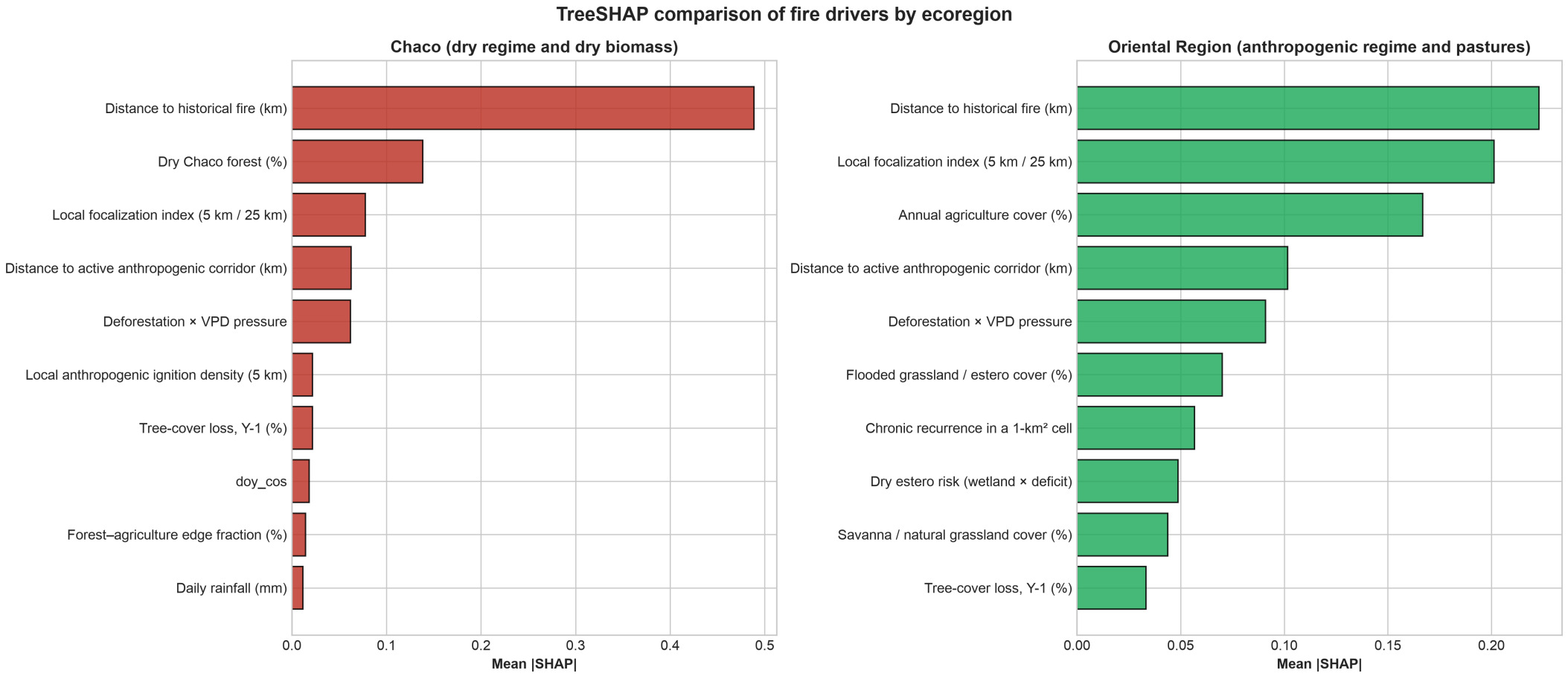


Figura S1. TreeSHAP comparison for the hybrid ignition model. The two regions use separate models and feature sets.

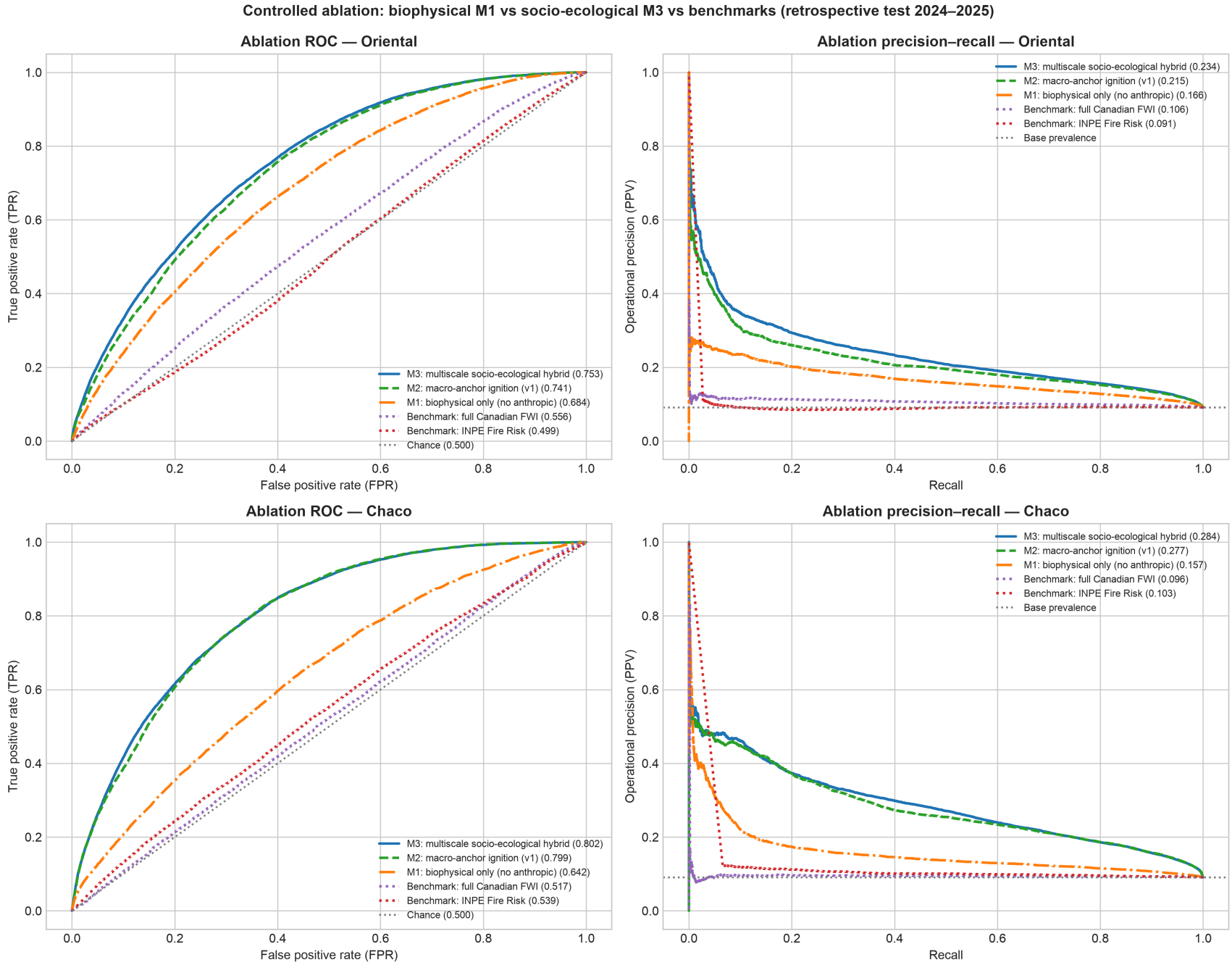


Figura S2. Controlled ablation under the D2 test. The figure complements Table S12.

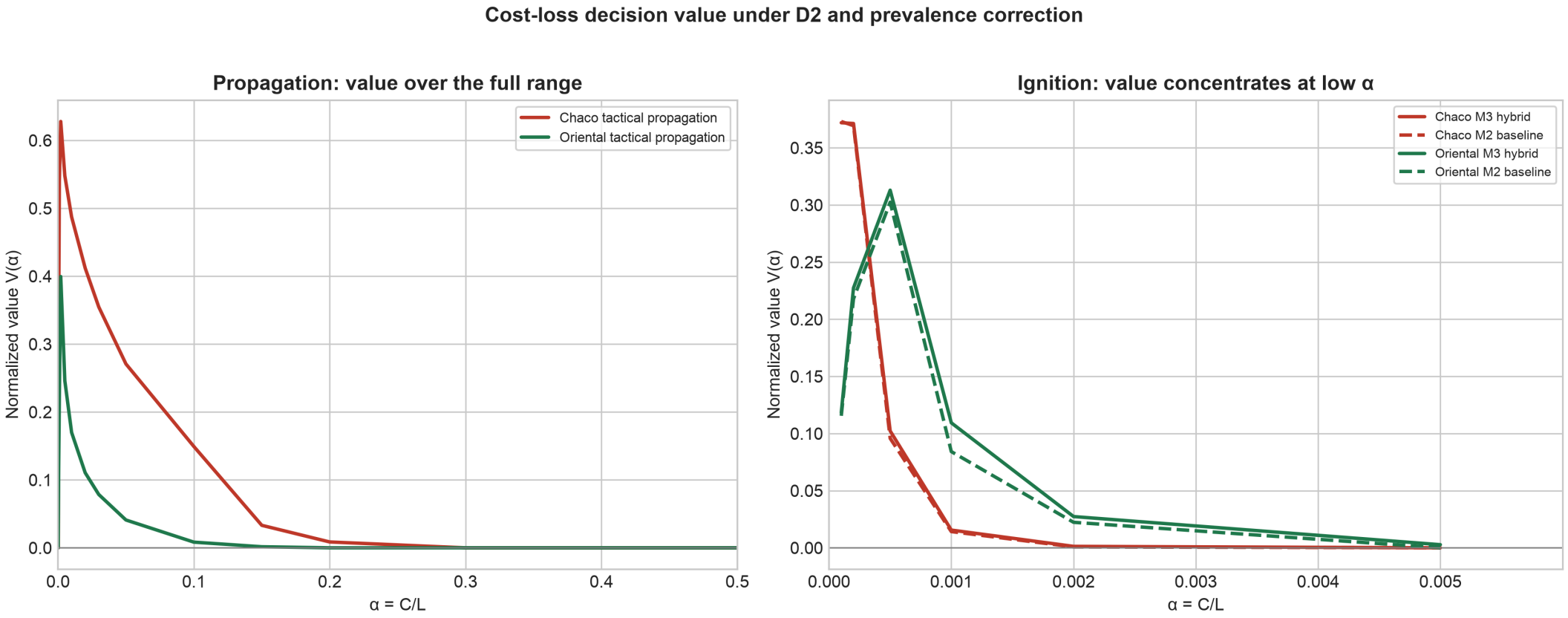


Figura S3. Cost-loss curves. The propagation panel uses the full alpha range; the ignition panel is truncated at alpha=0.006 because the decision value is zero or negligible at larger values. The figure complements Table S14.